

# Anmol Singh

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## Professional Summary

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Computer Science Engineer with hands-on experience in full-stack web development (MERN stack – MongoDB, Express.js, React.js, Node.js) and AI/ML, deep learning, and computer vision. Experienced in building and deploying machine learning systems using PyTorch, TensorFlow, FastAPI, React, and modern transformer architectures. Solved 200+ DSA problems across arrays, strings, linked lists, trees, matrices, and other core problem-solving topics.

## Technical Skills

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**Languages:** C/C++, Python, JavaScript, SQL, HTML, CSS

**Machine Learning:** PyTorch, TensorFlow, timm

**Backend:** FastAPI, Express.js, REST APIs

**Frontend:** React.js, Redux, RTK Query, Tailwind CSS

**Databases & Cloud:** MySQL, MongoDB, MongoDB Atlas, Cloudinary

**Tools & DevOps:** Git, GitHub, Postman, CI/CD

## Projects

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**MedScan AI – Full-Stack Medical Imaging Diagnosis Platform** — GitHub

- Built a full-stack web application integrating four CNN models to analyze X-ray, MRI, and CT scans for tuberculosis, brain tumors, lung cancer, and bone fractures (wrist, elbow, hand, foot).
- Implemented secure user authentication with JWT-based login/signup, along with a contact page, allowing users to upload scans, select the appropriate diagnostic model, and generate downloadable analysis reports.
- Designed a confidence-score reporting system displaying per-class prediction probabilities (e.g., fracture vs. normal), with two models achieving 99.00%+ accuracy and the remaining two achieving 98.00%+ accuracy.

**PlantDx – Fruit Plant Leaf Disease Classification Platform** — GitHub

- Developed a full-stack website leveraging four transformer-based models to classify leaf diseases across multiple fruit plants, including apple, mango, cherry, and grape.
- Enabled users to upload leaf images, select the appropriate classification model, and receive multi-class disease predictions with confidence scores, identifying whether the leaf is healthy or diseased.
- Implemented JWT-based user authentication and downloadable diagnostic reports summarizing prediction results and model confidence.

## Education

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**B.E. in Computer Engineering**

2024 – 2027

Thapar Institute of Engineering and Technology, Patiala

CGPA: 7.69